```
!pip install transformers torch pandas
!pip install transformers datasets accelerate
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Requirement already satisfied: transformers in
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Requirement already satisfied: MarkupSafe>=2.0 in
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
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Requirement already satisfied: mdurl~=0.1 in
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(0.1.2)
```

```python
import pandas as pd
from transformers import pipeline

# ------------------------------
# Load your dataset
# ------------------------------
df = pd.read_csv("/content/final_2600_dataset.csv")

# ------------------------------
# Load sentiment model
# ------------------------------
classifier = pipeline("sentiment-analysis")

# ------------------------------
# Generate sentiments
# ------------------------------
# (handles all rows)
results = classifier(df["dialogue"].tolist())

# ------------------------------
# Add sentiment column
# ------------------------------
df["sentiment"] = [r["label"].upper() for r in results]

# ------------------------------
# Optional: Normalize labels to 3 classes
# ------------------------------
def normalize_sentiment(label):
    if "POS" in label:
        return "POSITIVE"
    elif "NEG" in label:
        return "NEGATIVE"
    else:
        return "NEUTRAL"

df["sentiment"] = df["sentiment"].apply(normalize_sentiment)

# ------------------------------
# Save updated dataset
# ------------------------------
df.to_csv("dataset_with_sentiment.csv", index=False)

# ------------------------------
# Show result
# ------------------------------
```

```
print("Done □")
print("Shape:", df.shape)
df.head()
```

```
No model was supplied, defaulted to distilbert/distilbert-base-
uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in
production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py
:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```


```
{"model_id":"e647f976e510456ca1b1fa25a0807531","version_major":2,"vers
ion_minor":0}

{"model_id":"50d12686e7c046d1a5f42c9a15e2a8ac","version_major":2,"vers
ion_minor":0}

{"model_id":"1801b1246a0c4eed8f0b20ea1e48c214","version_major":2,"vers
ion_minor":0}

{"model_id":"c849dd6d681141808af3190db505a258","version_major":2,"vers
ion_minor":0}

{"model_id":"f34023f7e5e343fb85920b60cbb44a60","version_major":2,"vers
ion_minor":0}
```


```
Done □
Shape: (2600, 3)
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 2600,\n  \"fields\":
[\n    {\n      \"column\": \"character\",\n      \"properties\": {\n
\"dtype\": \"category\",\n        \"num_unique_values\": 91,\n
\"samples\": [\n          \"ANDREWS\",\n          \"OLAF\",\n
\"SIR COSMO\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"dialogue\",\n      \"properties\": {\n        \"dtype\":
\"string\",\n        \"num_unique_values\": 2596,\n
```

\"samples\": [\n   \"Bobby, we're launching. See these submersibles here, going in the water?\",\n   \"Oy, mate... that was a close shave.\",\n   \"closes his eyes, letting the chill wind clear his head.\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"sentiment\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 2,\n   \"samples\": [\n   \"NEGATIVE\",\n   \"POSITIVE\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe","variable_name":"df"}

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# ------------------------------
# Load dataset with sentiment
# ------------------------------
df = pd.read_csv("dataset_with_sentiment.csv")

# ------------------------------
# Encode labels
# ------------------------------
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

# ------------------------------
# Save updated dataset
# ------------------------------
df.to_csv("dataset_encoded.csv", index=False)

# ------------------------------
# Check mapping
# ------------------------------
print(dict(zip(le.classes_, le.transform(le.classes_))))
df.head()
```

```
{'NEGATIVE': np.int64(0), 'POSITIVE': np.int64(1)}
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 2600,\n \"fields\": [\n   {\n   \"column\": \"character\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 91,\n   \"samples\": [\n   \"ANDREWS\",\n   \"OLAF\",\n   \"SIR COSMO\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"dialogue\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 2596,\n   \"samples\": [\n   \"Bobby, we're launching. See these submersibles here, going in the water?\",\n   \"Oy, mate... that was a close shave.\",\n   \"closes his eyes, letting the chill wind clear his head.\"\n   ],\n   \"semantic_type\":

\"\",\n   \"description\": \"\"\n   }\n   },\n   {\n
\"column\": \"sentiment\",\n   \"properties\": {\n
\"dtype\": \"category\",\n   \"num_unique_values\": 2,\n
\"samples\": [\n   \"NEGATIVE\",\n   \"POSITIVE\"\n
],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
}\n   },\n   {\n   \"column\": \"label\",\n   \"properties\":
{\n   \"dtype\": \"number\",\n   \"std\": 0,\n
\"min\": 0,\n   \"max\": 1,\n   \"num_unique_values\": 2,\n
\"samples\": [\n   0,\n   1\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   }\n]\n}","type":"dataframe","variable_name":"df"}

```python
from sklearn.preprocessing import LabelEncoder

# check columns first
print(df.columns)

# create label column
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

# verify
print(df.head())
```

```
Index(['character', 'dialogue', 'sentiment', 'label'], dtype='object')
  character                                           dialogue
sentiment  \
0   ANATOLY                                       We are here.
POSITIVE
1   ANATOLY  5 MINUTES LATER: THE TWO SUBS skim over the se...
NEGATIVE
2   ANATOLY     sidescan sonar and the THRUM of big thrusters.
NEGATIVE
3   ANATOLY  6 The featureless gray clay of the bottom unro...
NEGATIVE
4   ANATOLY  Bodine is watching the sidescan sonar display,...
NEGATIVE

   label
0      1
1      0
2      0
3      0
4      0
```

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# ------------------------------
```

```python
# STEP 1: Load dataset
# ----------------------------
df = pd.read_csv("/content/dataset_with_sentiment.csv")

# ----------------------------
# STEP 2: Check columns
# ----------------------------
print("Columns:", df.columns)

# ----------------------------
# STEP 3: Fix column names (strip spaces)
# ----------------------------
df.columns = df.columns.str.strip()

# ----------------------------
# STEP 4: Check again
# ----------------------------
print("Fixed Columns:", df.columns)

# ----------------------------
# STEP 5: Ensure 'sentiment' exists
# ----------------------------
if "sentiment" not in df.columns:
    raise Exception("□ 'sentiment' column not found. Check file.")

# ----------------------------
# STEP 6: Create label column
# ----------------------------
le = LabelEncoder()
df["label"] = le.fit_transform(df["sentiment"])

print("Label column created □")

# ----------------------------
# STEP 7: Split dataset
# ----------------------------
X = df["dialogue"]
y = df["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Train size:", len(X_train))
print("Test size:", len(X_test))
```

```
Columns: Index(['character', 'dialogue', 'sentiment'], dtype='object')
Fixed Columns: Index(['character', 'dialogue', 'sentiment'],
dtype='object')
Label column created □
```

```
Train size: 2080
Test size: 520
```

```
from datasets import Dataset

# ------------------------------
# Convert pandas Series to Hugging Face Dataset format
# ------------------------------
train_dataset = Dataset.from_dict({
    "text": list(X_train),
    "label": list(y_train)
})

test_dataset = Dataset.from_dict({
    "text": list(X_test),
    "label": list(y_test)
})

print("Train and test datasets created □")
```

```
Train and test datasets created □
```

```
import numpy as np
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score, matthews_corrcoef, confusion_matrix,
mean_squared_error

def evaluate_model(y_true, y_pred, name):
    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred, average="weighted",
zero_division=0)
    recall = recall_score(y_true, y_pred, average="weighted",
zero_division=0)
    f1 = f1_score(y_true, y_pred, average="weighted", zero_division=0)
    mcc = matthews_corrcoef(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    cm = confusion_matrix(y_true, y_pred)

    print(f"\n{name} RESULTS")
    print("Accuracy:", accuracy)
    print("Precision:", precision)
    print("Recall:", recall)
    print("F1 Score:", f1)
    print("MCC:", mcc)
    print("RMSE:", rmse)
    print("Confusion Matrix:\n", cm)
```

```
!pip install transformers datasets accelerate
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
```

```
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: accelerate in
/usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.10.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: psutil in
/usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from accelerate)
(2.10.0+cu128)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.5)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in
```

```
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2026.2.25)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
```

```
>accelerate) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4-
>torch>=2.0.0->accelerate) (1.5.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2026.1)
```

```
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.23.0)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.13.0)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas->datasets) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (8.3.2)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-
hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.0.0-
>accelerate) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>huggingface-hub<2.0,>=1.3.0->transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers)
(0.1.2)
```

ELECTRA Model Training

```
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, Trainer, TrainingArguments,
DataCollatorWithPadding

# ------------------------------
# Load ELECTRA Tokenizer
# ------------------------------
model_name_electra = "google/electra-base-discriminator"
tokenizer_electra = AutoTokenizer.from_pretrained(model_name_electra)

def tokenize_electra(example):
    return tokenizer_electra(example["text"], truncation=True,
padding="max_length")

train_dataset_electra = train_dataset.map(tokenize_electra,
batched=True)
test_dataset_electra = test_dataset.map(tokenize_electra,
batched=True)

# ------------------------------
# Data collator
# ------------------------------
data_collator_electra =
DataCollatorWithPadding(tokenizer=tokenizer_electra)

print("ELECTRA tokenizer and data collator loaded □")
```


```
{"model_id":"5ff4d972b99e47a9b18c21bb254def4f","version_major":2,"vers
ion_minor":0}

{"model_id":"c795377c92024fed8fdc19a060ff79a1","version_major":2,"vers
ion_minor":0}

{"model_id":"43f3286af2ad4b92bfaebed8ce6c5bb3","version_major":2,"vers
ion_minor":0}
```

{"model_id":"5cbad64467c24f70936718f8a44bfe86","version_major":2,"version_minor":0}

{"model_id":"23db8e94c91d4e358b596c6b03338ccb","version_major":2,"version_minor":0}

{"model_id":"732abc106f114307b5471ee9686dc957","version_major":2,"version_minor":0}

```
ELECTRA tokenizer and data collator loaded 
```

ELECTRA with 3 Epochs

```
# ------------------------------
# Load ELECTRA model
# ------------------------------
model_electra_3 = AutoModelForSequenceClassification.from_pretrained(
    model_name_electra,
    num_labels=3
)

# ------------------------------
# Training arguments (3 epochs)
# ------------------------------
training_args_electra_3 = TrainingArguments(
    output_dir="./electra_results_3",
    num_train_epochs=3,
    per_device_train_batch_size=1,  # Further reduced batch size
    per_device_eval_batch_size=1,   # Further reduced batch size
    gradient_accumulation_steps=8,  # Accumulate gradients over 8
steps
    gradient_checkpointing=True,    # Enable gradient checkpointing
    logging_dir="./electra_logs_3",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_electra_3 = Trainer(
    model=model_electra_3,
    args=training_args_electra_3,
    train_dataset=train_dataset_electra,
    eval_dataset=test_dataset_electra,
    data_collator=data_collator_electra
)

# ------------------------------
# Train
# ------------------------------
```

```
trainer_electra_3.train()

# ----------------------------
# Predictions
# ----------------------------
y_pred_electra_3 =
trainer_electra_3.predict(test_dataset_electra).predictions.argmax(axi
s=1)

print("ELECTRA (3 epochs) DONE □")
```

{"model_id":"e6180f7448284ef2b743dd9bf4ca64b0","version_major":2,"vers
ion_minor":0}

{"model_id":"b9cf925a03b4460ca56d20b69e210a61","version_major":2,"vers
ion_minor":0}

```
ElectraForSequenceClassification LOAD REPORT from: google/electra-
base-discriminator
Key                                                  | Status     |
-----------------------------------------------------+------------+-
discriminator_predictions.dense.bias                 | UNEXPECTED |
discriminator_predictions.dense_prediction.weight    | UNEXPECTED |
electra.embeddings_project.bias                      | UNEXPECTED |
electra.embeddings_project.weight                    | UNEXPECTED |
discriminator_predictions.dense.weight               | UNEXPECTED |
discriminator_predictions.dense_prediction.bias      | UNEXPECTED |
classifier.out_proj.bias                             | MISSING    |
classifier.dense.bias                                | MISSING    |
classifier.out_proj.weight                           | MISSING    |
classifier.dense.weight                              | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.
```

{"model_id":"ac84b884bbcb48cebe5a4763e7bfcee7","version_major":2,"vers
ion_minor":0}

```
<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

ELECTRA (3 epochs) DONE □
```

ELECTRA with 5 Epochs

```
# ------------------------------
# Reload ELECTRA model (fresh training)
# ------------------------------
model_electra_5 = AutoModelForSequenceClassification.from_pretrained(
    model_name_electra,
    num_labels=3
)

# ------------------------------
# Training arguments (5 epochs)
# ------------------------------
training_args_electra_5 = TrainingArguments(
    output_dir="./electra_results_5",
    num_train_epochs=5,
    per_device_train_batch_size=1,  # Reduced batch size to 1
    per_device_eval_batch_size=1,   # Reduced batch size to 1
    gradient_accumulation_steps=8,  # Accumulate gradients over 8
steps
    gradient_checkpointing=True,    # Enable gradient checkpointing
    logging_dir="./electra_logs_5",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_electra_5 = Trainer(
    model=model_electra_5,
    args=training_args_electra_5,
    train_dataset=train_dataset_electra,
    eval_dataset=test_dataset_electra,
    data_collator=data_collator_electra
)

# ------------------------------
# Train
# ------------------------------
trainer_electra_5.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_electra_5 =
trainer_electra_5.predict(test_dataset_electra).predictions.argmax(axi
s=1)

print("ELECTRA (5 epochs) DONE □")
```

{"model_id":"4a0f80819cb9457f9f185a92aeb19f1e","version_major":2,"version_minor":0}

```
ElectraForSequenceClassification LOAD REPORT from: google/electra-base-discriminator
Key                                                  | Status     |
-----------------------------------------------------+------------+-
discriminator_predictions.dense.bias                 | UNEXPECTED |
discriminator_predictions.dense_prediction.weight    | UNEXPECTED |
electra.embeddings_project.bias                      | UNEXPECTED |
electra.embeddings_project.weight                    | UNEXPECTED |
discriminator_predictions.dense.weight               | UNEXPECTED |
discriminator_predictions.dense_prediction.bias      | UNEXPECTED |
classifier.out_proj.bias                             | MISSING    |
classifier.dense.bias                                | MISSING    |
classifier.out_proj.weight                           | MISSING    |
classifier.dense.weight                              | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

ELECTRA (5 epochs) DONE 
```

ELECTRA with 7 Epochs

```
# ------------------------------
# Reload ELECTRA model (fresh training)
# ------------------------------
model_electra_7 = AutoModelForSequenceClassification.from_pretrained(
    model_name_electra,
    num_labels=3
)

# ------------------------------
# Training arguments (7 epochs)
# ------------------------------
training_args_electra_7 = TrainingArguments(
    output_dir="./electra_results_7",
    num_train_epochs=7,
    per_device_train_batch_size=1,  # Reduced batch size to 1
    per_device_eval_batch_size=1,   # Reduced batch size to 1
    gradient_accumulation_steps=8,  # Accumulate gradients over 8
```

```
steps
    gradient_checkpointing=True,    # Enable gradient checkpointing
    fp16=True,                      # Enable mixed precision training
to save memory
    logging_dir="./electra_logs_7",
    save_strategy="no"
)

# ----------------------------
# Trainer
# ----------------------------
trainer_electra_7 = Trainer(
    model=model_electra_7,
    args=training_args_electra_7,
    train_dataset=train_dataset_electra,
    eval_dataset=test_dataset_electra,
    data_collator=data_collator_electra
)

# ----------------------------
# Train
# ----------------------------
trainer_electra_7.train()

# ----------------------------
# Predictions
# ----------------------------
y_pred_electra_7 =
trainer_electra_7.predict(test_dataset_electra).predictions.argmax(axi
s=1)

print("ELECTRA (7 epochs) DONE □")
```

{"model_id":"75d2d53e18154aaa97f23cff7ac223d2","version_major":2,"vers
ion_minor":0}

```
ElectraForSequenceClassification LOAD REPORT from: google/electra-
base-discriminator
Key                                                  | Status     |
-----------------------------------------------------+------------+-
discriminator_predictions.dense.bias                 | UNEXPECTED |
discriminator_predictions.dense_prediction.weight    | UNEXPECTED |
electra.embeddings_project.bias                      | UNEXPECTED |
electra.embeddings_project.weight                    | UNEXPECTED |
discriminator_predictions.dense.weight               | UNEXPECTED |
discriminator_predictions.dense_prediction.bias      | UNEXPECTED |
classifier.out_proj.bias                             | MISSING    |
classifier.dense.bias                                | MISSING    |
classifier.out_proj.weight                           | MISSING    |
classifier.dense.weight                              | MISSING    |
```

```
Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

ELECTRA (7 epochs) DONE 
```

ELECTRA Results

```
evaluate_model(y_test, y_pred_electra_3, "ELECTRA - 3 Epochs")
evaluate_model(y_test, y_pred_electra_5, "ELECTRA - 5 Epochs")
evaluate_model(y_test, y_pred_electra_7, "ELECTRA - 7 Epochs")
```

```
ELECTRA - 3 Epochs RESULTS
Accuracy: 0.8
Precision: 0.800803029611717
Recall: 0.8
F1 Score: 0.7989425618841625
MCC: 0.596783168199109
RMSE: 0.4472135954999579
Confusion Matrix:
 [[176  63]
 [ 41 240]]

ELECTRA - 5 Epochs RESULTS
Accuracy: 0.7865384615384615
Precision: 0.7872767187734293
Recall: 0.7865384615384615
F1 Score: 0.7853395322626091
MCC: 0.5694555257173589
RMSE: 0.4620189806290846
Confusion Matrix:
 [[172  67]
 [ 44 237]]

ELECTRA - 7 Epochs RESULTS
Accuracy: 0.8211538461538461
Precision: 0.8210370776150386
Recall: 0.8211538461538461
F1 Score: 0.8208473088455837
MCC: 0.6393332022912649
RMSE: 0.42290206176626033
```

```
Confusion Matrix:
 [[188  51]
 [ 42 239]]
```

RoBERTa Model Training (Re-run)

```
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, Trainer, TrainingArguments,
DataCollatorWithPadding
from datasets import Dataset

# ------------------------------
# Prepare datasets
# ------------------------------
train_dataset_roberta = Dataset.from_dict({
    "text": list(X_train),
    "label": list(y_train)
})

test_dataset_roberta = Dataset.from_dict({
    "text": list(X_test),
    "label": list(y_test)
})

# ------------------------------
# Load RoBERTa Tokenizer
# ------------------------------
model_name_roberta = "roberta-base"
tokenizer_roberta = AutoTokenizer.from_pretrained(model_name_roberta)

def tokenize_roberta(example):
    return tokenizer_roberta(example["text"], truncation=True,
padding="max_length")

train_dataset_roberta = train_dataset_roberta.map(tokenize_roberta,
batched=True)
test_dataset_roberta = test_dataset_roberta.map(tokenize_roberta,
batched=True)

# ------------------------------
# Data collator
# ------------------------------
data_collator_roberta =
DataCollatorWithPadding(tokenizer=tokenizer_roberta)

print("RoBERTa tokenizer and data collator loaded □")
```

{"model_id":"a2a1ca49124843868faf53871177cad6","version_major":2,"version_minor":0}

{"model_id":"63e4b4e31ba14f4a836d11490e5ec0c6","version_major":2,"version_minor":0}

{"model_id":"9d38a53d1a4a4a1d84173a6e0b8a3cb1","version_major":2,"version_minor":0}

{"model_id":"726ac0c7b2a6457dac48ef6503127f60","version_major":2,"version_minor":0}

{"model_id":"3e5a4f4b756946b084d54aafd604eb1a","version_major":2,"version_minor":0}

{"model_id":"51de7f2c282546cc88bad088238444d5","version_major":2,"version_minor":0}

{"model_id":"6beb9bd6a86049dfb70e6135089fc517","version_major":2,"version_minor":0}

```
RoBERTa tokenizer and data collator loaded 
```

RoBERTa with 3 Epochs

```
# ------------------------------
# Load RoBERTa model
# ------------------------------
model_roberta_3 = AutoModelForSequenceClassification.from_pretrained(
    model_name_roberta,
    num_labels=2 # Corrected from 3 to 2
)

# ------------------------------
# Training arguments (3 epochs)
# ------------------------------
training_args_roberta_3 = TrainingArguments(
    output_dir="./roberta_results_3",
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./roberta_logs_3",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_roberta_3 = Trainer(
    model=model_roberta_3,
    args=training_args_roberta_3,
    train_dataset=train_dataset_roberta,
    eval_dataset=test_dataset_roberta,
    data_collator=data_collator_roberta
```

```
)

# ----------------------------
# Train
# ----------------------------
trainer_roberta_3.train()

# ----------------------------
# Predictions
# ----------------------------
y_pred_roberta_3 =
trainer_roberta_3.predict(test_dataset_roberta).predictions.argmax(axi
s=1)

print("RoBERTa (3 epochs) DONE □")
```

{"model_id":"86f1438baee74f7db720b5142c5663b5","version_major":2,"vers
ion_minor":0}

```
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                              | Status     |
---------------------------------+------------+-
lm_head.dense.weight             | UNEXPECTED |
lm_head.bias                     | UNEXPECTED |
lm_head.dense.bias               | UNEXPECTED |
lm_head.layer_norm.bias          | UNEXPECTED |
lm_head.layer_norm.weight        | UNEXPECTED |
roberta.embeddings.position_ids  | UNEXPECTED |
classifier.out_proj.bias         | MISSING    |
classifier.dense.bias            | MISSING    |
classifier.out_proj.weight       | MISSING    |
classifier.dense.weight          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

RoBERTa (3 epochs) DONE □
```

RoBERTa with 5 Epochs

```
# ----------------------------
# Reload RoBERTa model (fresh training)
```

```
# ------------------------------
model_roberta_5 = AutoModelForSequenceClassification.from_pretrained(
    model_name_roberta,
    num_labels=2 # Corrected from 3 to 2
)

# ------------------------------
# Training arguments (5 epochs)
# ------------------------------
training_args_roberta_5 = TrainingArguments(
    output_dir="./roberta_results_5",
    num_train_epochs=5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./roberta_logs_5",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_roberta_5 = Trainer(
    model=model_roberta_5,
    args=training_args_roberta_5,
    train_dataset=train_dataset_roberta,
    eval_dataset=test_dataset_roberta,
    data_collator=data_collator_roberta
)

# ------------------------------
# Train
# ------------------------------
trainer_roberta_5.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_roberta_5 =
trainer_roberta_5.predict(test_dataset_roberta).predictions.argmax(axi
s=1)

print("RoBERTa (5 epochs) DONE □")
```

{"model_id":"85e49090dce94a7b8c27198ea25a85f6","version_major":2,"vers
ion_minor":0}

```
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                              | Status     |
---------------------------------+------------+-
lm_head.dense.weight             | UNEXPECTED |
```

```
lm_head.bias                      | UNEXPECTED |
lm_head.dense.bias                | UNEXPECTED |
lm_head.layer_norm.bias           | UNEXPECTED |
lm_head.layer_norm.weight         | UNEXPECTED |
roberta.embeddings.position_ids | UNEXPECTED |
classifier.out_proj.bias          | MISSING    |
classifier.dense.bias             | MISSING    |
classifier.out_proj.weight        | MISSING    |
classifier.dense.weight           | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

<IPython.core.display.HTML object>

---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call
last)
/tmp/ipykernel_621/1396081886.py in <cell line: 0>()
     33 # Train
     34 # ----------------------------
---> 35 trainer_roberta_5.train()
     36
     37 # ----------------------------

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
train(self, resume_from_checkpoint, trial, ignore_keys_for_eval)
   2172                 hf_hub_utils.enable_progress_bars()
   2173         else:
-> 2174             return inner_training_loop(
   2175                 args=args,
   2176                 resume_from_checkpoint=resume_from_checkpoint,

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_inner_training_loop(self, batch_size, args, resume_from_checkpoint,
trial, ignore_keys_for_eval)
   2539                         args.logging_nan_inf_filter
   2540                         and not is_torch_xla_available()
-> 2541                         and (torch.isnan(tr_loss_step) or
torch.isinf(tr_loss_step))
   2542                     ):
   2543                         # if loss is nan or inf simply add the
average of previous logged losses
```

```
KeyboardInterrupt:
```

RoBERTa with 7 Epochs

```
# ------------------------------
# Reload RoBERTa model (fresh training)
# ------------------------------
model_roberta_7 = AutoModelForSequenceClassification.from_pretrained(
    model_name_roberta,
    num_labels=2 # Corrected from 3 to 2
)

# ------------------------------
# Training arguments (7 epochs)
# ------------------------------
training_args_roberta_7 = TrainingArguments(
    output_dir="./roberta_results_7",
    num_train_epochs=7,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    logging_dir="./roberta_logs_7",
    save_strategy="no"
)

# ------------------------------
# Trainer
# ------------------------------
trainer_roberta_7 = Trainer(
    model=model_roberta_7,
    args=training_args_roberta_7,
    train_dataset=train_dataset_roberta,
    eval_dataset=test_dataset_roberta,
    data_collator=data_collator_roberta
)

# ------------------------------
# Train
# ------------------------------
trainer_roberta_7.train()

# ------------------------------
# Predictions
# ------------------------------
y_pred_roberta_7 =
trainer_roberta_7.predict(test_dataset_roberta).predictions.argmax(axi
s=1)

print("RoBERTa (7 epochs) DONE □")
```

{"model_id":"d497ea052e794836b3ab5c9fc30a0eb8","version_major":2,"version_minor":0}

```
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                                | Status     |
-----------------------------------+------------+-
lm_head.dense.weight               | UNEXPECTED |
lm_head.bias                       | UNEXPECTED |
lm_head.dense.bias                 | UNEXPECTED |
lm_head.layer_norm.bias            | UNEXPECTED |
lm_head.layer_norm.weight          | UNEXPECTED |
roberta.embeddings.position_ids    | UNEXPECTED |
classifier.out_proj.bias           | MISSING    |
classifier.dense.bias              | MISSING    |
classifier.out_proj.weight         | MISSING    |
classifier.dense.weight            | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
- MISSING  :those params were newly initialized because missing from
the checkpoint. Consider training on your downstream task.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

RoBERTa (7 epochs) DONE 
```

RoBERTa Results

```
evaluate_model(y_test, y_pred_roberta_3, "RoBERTa - 3 Epochs")
evaluate_model(y_test, y_pred_roberta_5, "RoBERTa - 5 Epochs")
evaluate_model(y_test, y_pred_roberta_7, "RoBERTa - 7 Epochs")
```

```
RoBERTa - 3 Epochs RESULTS
Accuracy: 0.801923076923077
Precision: 0.8017845423622281
Recall: 0.801923076923077
F1 Score: 0.8014915271095412
MCC: 0.6004169662903989
RMSE: 0.4450583367120799
Confusion Matrix:
 [[182  57]
 [ 46 235]]
```

```
RoBERTa - 5 Epochs RESULTS
Accuracy: 0.825
Precision: 0.824846841192995
Recall: 0.825
F1 Score: 0.8248466577923943
MCC: 0.6472717145094427
RMSE: 0.4183300132670378
Confusion Matrix:
 [[191  48]
 [ 43 238]]

RoBERTa - 7 Epochs RESULTS
Accuracy: 0.7865384615384615
Precision: 0.7870279645926709
Recall: 0.7865384615384615
F1 Score: 0.7854784887297995
MCC: 0.5693428069783585
RMSE: 0.4620189806290846
Confusion Matrix:
 [[173  66]
 [ 45 236]]
```

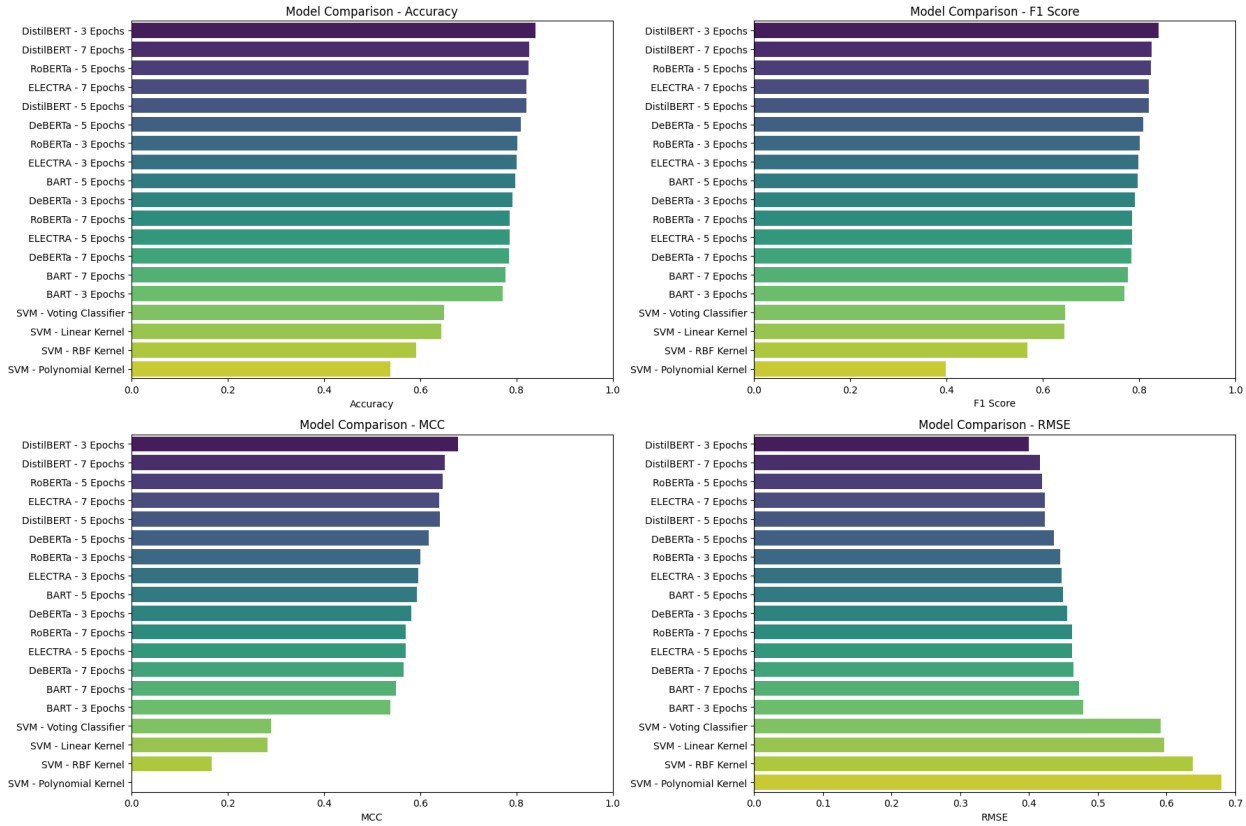
all models comparison

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Data for BART models
bart_data = {
    "Model": ["BART - 3 Epochs", "BART - 5 Epochs", "BART - 7
Epochs"],
    "Accuracy": [0.7711538461538462, 0.7980769230769231,
0.7769230769230769],
    "Precision": [0.7728735207100591, 0.7982894161585437,
0.7767527162103433],
    "Recall": [0.7711538461538462, 0.7980769230769231,
0.7769230769230769],
    "F1 Score": [0.7692001098109273, 0.7973181578854086,
0.7762686965811966],
    "MCC": [0.5387637541291109, 0.5926282567542644,
0.5497571051179371],
    "RMSE": [0.47837867202264966, 0.4493585171364586,
0.4723101979387308]
}

# Data for DeBERTa models
deberta_data = {
    "Model": ["DeBERTa - 3 Epochs", "DeBERTa - 5 Epochs", "DeBERTa - 7
```

```
Epochs"],
    "Accuracy": [0.7923076923076923, 0.8096153846153846,
0.7846153846153846],
    "Precision": [0.7943420431320245, 0.8122904339250493,
0.7854599841637585],
    "Recall": [0.7923076923076923, 0.8096153846153846,
0.7846153846153846],
    "F1 Score": [0.7906161828036827, 0.8079900073216958,
0.7833330056451044],
    "MCC": [0.581941864211242, 0.6175303848497411,
0.5656075007013708],
    "RMSE": [0.45573271518765, 0.4363308554120547, 0.4640954808922571]
}

# Data for DistilBERT models
distilbert_data = {
    "Model": ["DistilBERT - 3 Epochs", "DistilBERT - 5 Epochs",
"DistilBERT - 7 Epochs"],
    "Accuracy": [0.8403846153846154, 0.8211538461538461,
0.8269230769230769],
    "Precision": [0.8407813892188891, 0.8236296230651069,
0.8268009383926338],
    "Recall": [0.8403846153846154, 0.8211538461538461,
0.8269230769230769],
    "F1 Score": [0.8398737763690544, 0.8197661252520896,
0.826664656731846],
    "MCC": [0.6783142721582107, 0.6407901971919611,
0.6510132237896609],
    "RMSE": [0.3995189414976274, 0.42290206176626033,
0.41602514716892186]
}

# Data for SVM models
svm_data = {
    "Model": ["SVM - Linear Kernel", "SVM - RBF Kernel", "SVM -
Polynomial Kernel", "SVM - Voting Classifier"],
    "Accuracy": [0.6442307692307693, 0.5923076923076923,
0.5384615384615384, 0.65],
    "Precision": [0.6441216011772958, 0.5943856143856143,
0.5041723562433622, 0.6489091078615717],
    "Recall": [0.6442307692307693, 0.5923076923076923,
0.5384615384615384, 0.65],
    "F1 Score": [0.64417380397306, 0.5674550321609145,
0.3976869154534129, 0.6462817233916979],
    "MCC": [0.2835673204423471, 0.1668043104426767 4,
0.0006178962877914625, 0.29027984987504657],
    "RMSE": [0.596463939202724, 0.6385078759829887,
0.6793662204867574, 0.5916079783099616]
}
```

```
# Data for RoBERTa models
roberta_data = {
    "Model": ["RoBERTa - 3 Epochs", "RoBERTa - 5 Epochs", "RoBERTa - 7
Epochs"],
    "Accuracy": [0.801923076923077, 0.825, 0.7865384615384615],
    "Precision": [0.8017845423622281, 0.824846841192995,
0.7870279645926709],
    "Recall": [0.801923076923077, 0.825, 0.7865384615384615],
    "F1 Score": [0.8014915271095412, 0.8248466577923943,
0.7854784887297995],
    "MCC": [0.6004169662903989, 0.6472717145094427,
0.5693428069783585],
    "RMSE": [0.4450583367120799, 0.4183300132670378,
0.4620189806290846]
}

# Data for ELECTRA models
electra_data = {
    "Model": ["ELECTRA - 3 Epochs", "ELECTRA - 5 Epochs", "ELECTRA - 7
Epochs"],
    "Accuracy": [0.8, 0.7865384615384615, 0.8211538461538461],
    "Precision": [0.800803029611717, 0.7872767187734293,
0.8210370776150386],
    "Recall": [0.8, 0.7865384615384615, 0.8211538461538461],
    "F1 Score": [0.7989425618841625, 0.7853395322626091,
0.8208473088455837],
    "MCC": [0.596783168199109, 0.5694555257173589,
0.6393332022912649],
    "RMSE": [0.4472135954999579, 0.4620189806290846,
0.42290206176626033]
}

# Create DataFrames
df_bart = pd.DataFrame(bart_data)
df_deberta = pd.DataFrame(deberta_data)
df_distilbert = pd.DataFrame(distilbert_data)
df_svm = pd.DataFrame(svm_data)
df_roberta = pd.DataFrame(roberta_data)
df_electra = pd.DataFrame(electra_data)

# Combine all results
all_results = pd.concat(
    [df_bart, df_deberta, df_distilbert, df_svm, df_roberta,
df_electra],
    ignore_index=True
)

# Sort by Accuracy
all_results_sorted = all_results.sort_values(by="Accuracy",
ascending=False)
```

```python
# Melt for plotting
all_results_melted = all_results_sorted.melt(id_vars="Model",
var_name="Metric", value_name="Score")

# Metrics to plot
metrics_to_plot = ["Accuracy", "F1 Score", "MCC", "RMSE"]
plot_data =
all_results_melted[all_results_melted["Metric"].isin(metrics_to_plot)]

# Plot
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(18, 12))
axes = axes.flatten()

for i, metric in enumerate(metrics_to_plot):
    sns.barplot(
        x="Score",
        y="Model",
        hue="Model",
        data=plot_data[plot_data["Metric"] == metric],
        ax=axes[i],
        palette="viridis",
        legend=False
    )
    axes[i].set_title(f"Model Comparison - {metric}")
    axes[i].set_xlabel(metric)
    axes[i].set_ylabel("")
    axes[i].set_xlim(0, 1)

    if metric == "RMSE":
        axes[i].set_xlim(0, 0.7)

plt.tight_layout()
plt.show()

print("Model comparison graphs generated □")
```




```
Model comparison graphs generated 
```